Utilizing GitHub Copilot

Steve Buchanan
Principal Program Manager

@buchatech | www.buchatech.com

Overview

Integrating Copilot Into IaC

Exploring Copilot in VS Code

Using Various Parts of Functionality of GitHub Copilot

Integrating Copilot Into IaC

IaC in DevOps

Infrastructure as Code (IaC) helps DevOps teams in various ways:

- Swiftly establish comprehensive environments spanning from development to production
- Safeguard consistent and reproducible configurations across different environments
- Seamlessly integrate with cloud providers and optimize the scaling of infrastructure resources according to demand
- IaC establishes a shared language for both developers and operations. This facilitates transparent review processes for changes, fostering improved collaboration within a DevOps environment

AI holds the potential to streamline infra deployment by automating numerous intricate & labor-intensive tasks associated with Infra as Code (IaC)

Uses for GitHub Copilot in IaC

IaC Generation - GitHub Copilot is capable of producing scripts or templates for IaC that provisions & configures cloud resources, including servers, containers, networks, storage, functions & more.

CI/CD Pipeline Generation - GitHub Copilot is capable of producing Continuous Integration and Continuous Delivery (CI/CD) pipelines to automate the testing, building, and deployment processes for Infrastructure as Code (IaC). i.e. Copilot could generate a GitHub Actions workflow executing unit tests & deploys IaC to Azure/AWS/GCP.

Policy as Code Generation - GitHub Copilot is capable of generating Policy as Code (PaC) scripts/templates for enforcing rules & compliance on cloud resources. i.e.: it can create Azure Policies for maintaining compliance in a virtual network.

Configuration File Generation - GitHub Copilot is able to generate config files that specify settings & parameters for various apps, frameworks, or tools. i.e. a Copilot prompt such as: "Generate a K8 ConfigMap file with “abc” settings to update the config.properties file in a Java app."

IaC languages that work with GitHub Copilot

Terraform (Hashicorp Configuration Language - HCL) – Cloud etc...

Bash – Linux, Containers, Cloud

PowerShell – Windows, Cloud

Python – Linux, Windows, Cloud, Pulumi

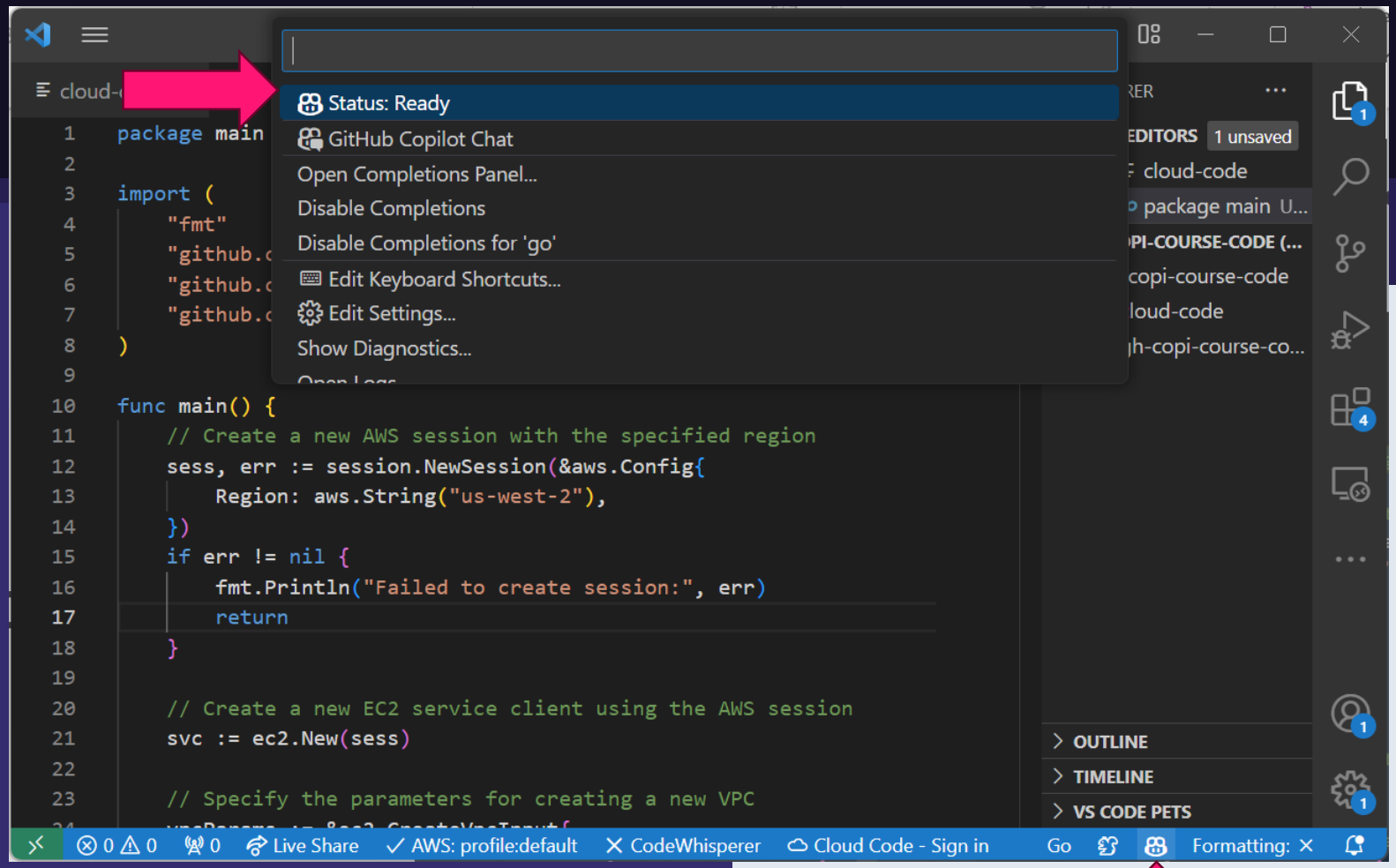
YAML – Kubernetes, Crossplane

Exploring Copilot in VS Code

Copilot Menu in VS Code

Github Copilot Icon indicates whether GitHub Copilot is enabled or disabled

GitHub Copilot Options & Access to Configuration



Copilot Config in VS Code

Configuration

File Edit Selection View Go Run Terminal Help

gh-copi-course-code (Workspace)

cloud-code

Settings

GitHub Copilot

12 Settings Found

User Workspace gh-copi-course-code Folder

Last synced: 8 mos ago

Text Editor (2)
Extensions (10)
Code GPT: Chat &... (1)
CodeGPT Auto... (1)
Copilot (3)
Copilot (4)
GitHub Copilot V... (2)

Editor: Default Folding Range Provider

Defines a default folding range provider that takes precedence over all other folding range providers. Must be the identifier of an extension contributing a folding range provider.

All

Editor: Default Formatter

Defines a default formatter which takes precedence over all other formatter settings. Must be the identifier of an extension contributing a formatter.

None

Code GPT.Autocomplete: Enabled

☑ Enable CodeGPT Copilot

Github › Copilot: Advanced

Edit in settings.json

Github › Copilot › Editor: Enable Auto Completions

☑ Automatically show inline completions

Github › Copilot: Enable

Enable or disable Copilot completions for specified languages

Item	Value
*	true
plaintext	false
markdown	false
scminput	false

Add Item

Github › Copilot › Chat: Locale Override

Specify a locale that Copilot should respond in, e.g. `en` or `fr`. By default, Copilot will respond using VS Code's configured display language locale.

auto

GitHub Copilot Keyboard Shortcuts

ESC Dismiss suggestion	**TAB** Accept suggestion	**ALT+]** Next suggestion
ALT+[Previous suggestion	**CTRL+ENTER** See 10 suggestions	**Alt + ** Trigger suggestion

Generating suggestions from comments

Describe what you want in VS Code within a comment, then GitHub Copilot provide code suggestions to help you achieve your objective

```
# please give me the code in YAML to deploy a WordPress container to a Kubernetes cluster

apiVersion: apps/v1
kind: Deployment
metadata:
  name: wordpress
spec:
  selector:
    matchLabels:
      app: wordpress
      replicas: 1
        template:
    metadata:
      labels:
        app: wordpress
    spec:
      containers:
      - name: wordpress
        image: wordpress:4.8-apache
        ports:
        - containerPort: 80
        env:
        - name: WORDPRESS_DB_HOST
          value: wordpress-mysql
        - name: WORDPRESS_DB_PASSWORD
          valueFrom:
            secretKeyRef:
              name: mysql-pass
              key: password
        volumeMounts:
        - name: wordpress-persistent-storage
          mountPath: /var/www/html
……
……
……
```

Chat-GPT Chat

Two ways to use AI chat inside of VS Code

File Edit Selection View Go ···

gh-copi-course-code (Workspace)

cloud-code | Untitled-2 | package main Untitled-1

Ask Copilot or type '/' for commands (Ctrl+UpArrow, Ctrl+DownArrow for history)

Copilot generated code may be incorrect

1 *Press Ctrl+I to ask GitHub Copilot Chat to do something. Start typir to dismiss.*

CODEGPT CHAT

GPT-4 Free Trial!
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Get Started for Free

CODEGPT

Hi, I'm CodeGPT by **Judini**, your coding sidekick!. Feel free to ask me any coding related questions.

To get started, simply select a section of code and choose one of the following options:

- Explain the selected code.
- Identify any issues in my selected code.
- Create unit tests for my selected code.

If you want to learn more about me, check out the Documentation

Big news 🤯! Don't miss Judini's waiting list. Soon, train CodeGPT with your data (eg. your repos!) and create your own AI Agents 🤖

Settings

Send message...

Document Code

Highlight code and then ask Copilot to document code via chat

Please document this code

Accept | Discard | Changed 57 lines

```go
import (
    "fmt"
    "github.com/aws/aws-sdk-go/aws"
    "github.com/aws/aws-sdk-go/aws/session"
    "github.com/aws/aws-sdk-go/service/ec2"
)

func main() {

    sess, err := session.NewSession(&a
        Region: aws.String("us-west-2"
    })
    if err != nil {
        fmt.Println("Failed to create
        return
    }

    svc := ec2.New(sess)

    vpcParams := &ec2.CreateVpcInput{
        CidrBlock: aws.String("10.0.0.
    }

    result, err := svc.CreateVpc(vpcPa
    if err != nil {
        fmt.Println("Failed to create
        return
    }
```

```go
// main is the entry point of the program.
func main() {

    // Create a new AWS session with the specified region.
    sess, err := session.NewSession(&aws.Config{
        Region: aws.String("us-west-2"),
    })
    if err != nil {
        fmt.Println("Failed to create session:", err)
        return
    }

    // Create a new EC2 service client using the session.
    svc := ec2.New(sess)

    // Specify the parameters for creating a new VPC.
    vpcParams := &ec2.CreateVpcInput{
        CidrBlock: aws.String("10.0.0.0/16"),
    }

    // Create a new VPC using the specified parameters.
    result, err := svc.CreateVpc(vpcParams)
    if err != nil {
        fmt.Println("Failed to create VPC:", err)
        return
    }
```

Using Various Parts of Functionality of GitHub Copilot

Demo

Demo: Use GitHub Copilot Comment-based Suggestions

Demo

Demo: Use GitHub Copilot to document my code

Demo

Demo: Use GitHub Copilot to explain existing code

Demo

Demo: Learn a New Language with GitHub Copilot

Summary

In this module we covered:

- Integrating Copilot Into IaC
- Exploring Copilot in VS Code
- Demos of various functionality of GitHub Copilot

Why this is important:?

- This module is core to your knowledge learning how to use GitHub Copilot so you can get started using Copilot to assist you with your IaC.

Up Next:

Real-world Applications and Conclusion